

Conduct a comprehensive literature review on 'Cancer-Related Cognitive Impairment (CRCI) in lung cancer patients undergoing targeted therapy'. please cover Clinical Burden & Prevalence Summarize the current evidence regarding the prevalence, severity, and impact of cognitive/emotional dysfunction on the quality of life and social reintegration of lung cancer survivors. Mechanistic Insights: Analyze how targeted therapies (e.g., EGFR-TKIs, ALK inhibitors) potentially induce neurotoxicity, specifically focusing on structural and functional brain changes (e.g., neuroinflammation, blood-brain barrier disruption, or synaptic plasticity impairment). Current Management Gaps: Identify the limitations of existing clinical management for CRCI in lung cancer patients and highlight the lack of standardized rehabilitation protocols.

While cancer-related cognitive impairment affects approximately one-quarter of lung cancer survivors and impacts memory, processing speed, attention, and quality of life, the existing literature derives predominantly from chemotherapy studies in other cancer types and provides virtually no specific evidence regarding prevalence, mechanisms, or management of cognitive impairment in patients receiving EGFR-TKIs or ALK inhibitors, revealing a critical knowledge gap despite the clinical importance of this issue.

Abstract

Cancer-related cognitive impairment affects approximately one-quarter of lung cancer survivors, with pooled prevalence of 26% (95% PI, 16–37%) [1], though regional variation is substantial, ranging from 15% in Asia to 44% in North America [1]. Memory, processing speed, attention, and executive functions are the most commonly impaired domains [2, 2], significantly impacting quality of life [1] and functional status including activities of daily living [3]. However, critical knowledge gaps exist regarding CRCI in patients receiving targeted therapies: no studies provided specific prevalence data, severity classifications, or temporal patterns for EGFR-TKI or ALK inhibitor recipients [1–5]. Mechanistic evidence suggests chronic neuroinflammation with elevated pro-inflammatory cytokines [2, 4], blood-brain barrier disruption [3], mitochondrial dysfunction [2, 3], altered Default Mode Network connectivity [3], and grey matter volume reductions [4], but these findings derive predominantly from chemotherapy studies in breast cancer populations [2, 2, 2] rather than targeted therapy in lung cancer. Management is hampered by the absence of standardized assessment tools [1, 4], evidence-based treatment guidelines [1, 2], and rehabilitation programs [2], with non-pharmacological interventions including cognitive training and exercise representing first-line approaches [2, 4] despite limited evidence in lung cancer populations. The literature reveals a paradox: while CRCI is recognized as prevalent and impactful in lung cancer survivors, the specific contribution, mechanisms, and optimal management of targeted therapy-induced cognitive impairment remain poorly characterized.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "Conduct a comprehensive literature review on 'Cancer-Related Cognitive Impairment (CRCI) in lung cancer patients undergoing targeted therapy'. please cover Clinical Burden & Prevalence Summarize the current evidence regarding the prevalence, severity, and impact of cognitive/emotional dysfunction on the quality of life and social reintegration of lung cancer survivors. Mechanistic Insights: Analyze how targeted therapies (e.g., EGFR-TKIs, ALK inhibitors) potentially induce neurotoxicity, specifically focusing on structural and functional brain changes (e.g., neuroinflammation, blood-brain barrier disruption, or synaptic plasticity impairment). Current Management Gaps: Identify the limitations of existing clinical management for CRCI in lung cancer patients and highlight the lack of standardized rehabilitation protocols."

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Lung Cancer Population:** Does this study include patients diagnosed with lung cancer (any stage)?

- **Targeted Therapy Intervention:** Does this study investigate patients receiving targeted therapeutic agents (such as EGFR-TKIs, ALK inhibitors, ROS1 inhibitors, BRAF inhibitors, MET inhibitors, RET inhibitors, or other molecularly targeted therapies)?
- **Cognitive Assessment Outcomes:** Does this study measure cognitive function, cognitive impairment, neurocognitive changes, or cancer-related cognitive impairment using validated neuropsychological tests, cognitive screening tools, or patient-reported cognitive outcomes?
- **Lung Cancer Focus:** If this study includes multiple cancer types, does it provide separate analysis or results specifically for lung cancer patients?
- **Targeted Therapy Component:** Does this study include targeted therapy as a component (not exclusively chemotherapy, radiation therapy, immunotherapy, or surgery alone)?
- **Cognitive Effects Attribution:** Are the cognitive effects in this study potentially attributable to targeted therapy (rather than being primarily attributed to brain metastases, paraneoplastic syndromes, or pre-existing neurological conditions unrelated to treatment)?
- **Human Clinical Study:** Is this a study conducted in human participants (not exclusively animal studies, in vitro studies, or purely laboratory-based research)?
- **Adequate Study Quality:** Is this study either: (1) a full publication (not just a conference abstract), AND (2) if it is a case series, does it include 5 or more participants?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** n = 45

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Population:**

Extract details about the lung cancer patient population, including:

- Cancer stage and histology (NSCLC vs SCLC)
- Sample size of lung cancer patients specifically
- Age, gender, and baseline cognitive status
- Presence of brain metastases at baseline
- Time since diagnosis and treatment status
- Any genetic markers (EGFR, ALK, etc.) if reported

- **Targeted Therapy:**

Extract specific information about targeted therapies used in lung cancer patients, including:

- Type of targeted agent (EGFR-TKIs, ALK inhibitors, other)
- Specific drug names and generations (1st, 2nd, 3rd gen)
- Dosage, duration, and treatment schedule
- CNS penetration properties if mentioned
- Combination with other treatments
- Treatment line (first-line, subsequent)

- **CRCI Assessment:**

Extract methods used to assess cognitive impairment in lung cancer patients, including:

- Assessment tools used (neuropsychological tests, self-report measures, screening tools)
- Specific cognitive domains evaluated (memory, attention, executive function, processing speed)
- Timing of assessment relative to treatment initiation
- Use of ICCTF criteria or other standardized definitions
- Baseline vs. post-treatment assessments

- **CRCI Prevalence:**

Extract quantitative data on cognitive impairment prevalence and severity in lung cancer patients on targeted therapy, including:

- Overall prevalence rates with confidence intervals
- Severity classifications (mild, moderate, severe)
- Incidence of new-onset cognitive impairment
- Progression or recovery rates over time
- Comparison rates between different targeted therapies if available

- **Quality of Life Impact:**

Extract data on how CRCI affects quality of life and social functioning in lung cancer patients, including:

- Quality of life measures and scores
- Impact on activities of daily living and instrumental activities
- Social reintegration outcomes (return to work, social relationships)
- Functional status and independence measures
- Patient-reported burden and distress related to cognitive symptoms
- Caregiver burden if assessed

- **Mechanistic Findings:**

Extract evidence of mechanisms by which targeted therapies cause cognitive impairment, including:

- Neuroinflammation markers and pathways
- Blood-brain barrier disruption evidence
- Synaptic plasticity changes
- Neurotransmitter system alterations
- DNA damage or mitochondrial dysfunction
- Immune system involvement
- Direct neurotoxicity mechanisms
- Any biomarkers of neurotoxicity identified

- **Brain Changes:**

Extract structural and functional brain changes associated with targeted therapy in lung cancer patients, including:

- Neuroimaging findings (MRI, fMRI, PET)
- Gray and white matter volume changes
- Connectivity alterations (default mode network, others)

- Hippocampal or frontal lobe changes specifically
- Functional activation patterns
- Correlation between brain changes and cognitive symptoms

- **Current Management:**

Extract information about current clinical management approaches for CRCI in lung cancer patients, including:

- Screening and monitoring practices
- Pharmacological interventions used
- Non-pharmacological interventions (cognitive training, exercise)
- Rehabilitation protocols or programs
- Multidisciplinary care approaches
- Effectiveness outcomes of management strategies
- Healthcare provider involvement (neurologists, neuropsychologists, etc.)

- **Management Gaps:**

Extract identified limitations and gaps in CRCI management for lung cancer patients, including:

- Lack of standardized assessment protocols
- Absence of evidence-based treatment guidelines
- Limited rehabilitation program availability
- Barriers to implementation of interventions
- Unmet patient needs identified
- Recommendations for improved care delivery
- Research priorities highlighted by authors

Results

Characteristics of Included Studies

The review included 5 sources examining cancer-related cognitive impairment in lung cancer patients, with varying availability of full-text access and methodological approaches.

Study	Full text retrieved?	Study Type	Population Focus	Key Focus Areas
Z. Orszaghova et al., 2021 [4]	Yes	Narrative review	Multiple cancer types including lung cancer [4]	Long-term CRCI effects, mechanisms, biomarkers [4]
A. Wrona et al., 2018 [5]	No	Narrative review	NSCLC with EGFR and ALK alterations [5]	Brain metastases management, targeted therapies [5]
Murphy Ho et al., 2024 [1]	Yes	Systematic review and meta-analysis	1934 lung cancer survivors across 12 studies [1]	CRCI prevalence and assessment tools [1]

Study	Full text retrieved?	Study Type	Population Focus	Key Focus Areas
M. Lange et al., 2019 [2]	Yes	Narrative review	Multiple cancer types, primarily breast cancer [2]	CRCI detection and management strategies [2]
Tianxue Hou et al., 2024 [3]	No	Systematic review	17 observational studies on lung cancer patients [3]	Factors associated with cognitive impairment [3]

The included studies varied substantially in their focus and methodology. Two studies specifically targeted lung cancer populations [1, 3], while three examined CRCI across multiple cancer types with varying degrees of lung cancer inclusion [2, 4, 5]. Among the lung cancer-specific studies, Murphy Ho et al. analyzed data from 1934 survivors with mean ages ranging from 59.1 to 68.8 years and male participant ratios from 46% to 100% [1]. Only one study specifically addressed NSCLC patients with oncogene-driven disease, focusing on EGFR-mutated and ALK-rearranged subtypes [5, 5]. However, none of the sources provided detailed information about cancer staging, treatment status, presence of brain metastases at baseline, or specific genetic marker distributions in their lung cancer cohorts [1–5].

Regarding targeted therapy details, the evidence was notably limited. While EGFR-TKIs and ALK inhibitors were mentioned as relevant agents [5], none of the sources provided specific information about drug names, generations (first, second, or third), dosages, treatment duration, CNS penetration properties, combination regimens, or treatment line positioning [1–5]. This represents a substantial gap in characterizing the relationship between specific targeted therapy regimens and cognitive outcomes.

Clinical Burden and Prevalence

Prevalence of CRCI in Lung Cancer Survivors

The pooled prevalence of CRCI in lung cancer survivors was 26% (95% PI, 16–37%) [1], though this estimate showed substantial heterogeneity ($I^2 = 95.97\%$) [1]. Regional variation was pronounced, with prevalence rates of 15% (95% PI, 8–25%) in Asia, 33% (95% PI, 13–57%) in Europe, and 44% (95% PI, 37–52%) in North America [1]. The region where studies were conducted was identified as a significant factor contributing to this heterogeneity ($p = 0.013$) [1].

Assessment methodology also influenced prevalence estimates substantially. Studies using objective comprehensive neuropsychological evaluations reported a prevalence of 36% (95% PI, 16–58%), while those using objective screening tools alone found 17% (95% PI, 7–30%), and subjective self-report measures identified CRCI in 22% (95% PI, 10–37%) of survivors [1]. For comparison, in breast cancer populations—the most extensively studied cancer type—50% of patients reported cognitive complaints while only 15–25% demonstrated objective cognitive decline on formal testing [2]. This discordance between subjective complaints and objective performance appears to be a consistent feature across cancer types.

The available evidence did not provide specific prevalence data for lung cancer patients receiving targeted therapies, severity classifications (mild, moderate, severe), incidence of new-onset cognitive impairment, or progression and recovery rates over time [1–5]. This represents a critical knowledge gap for counseling patients initiating EGFR-TKIs or ALK inhibitors.

Cognitive Domains Affected

Memory, processing speed, attention, and executive functions emerged as the most commonly impaired domains following cancer treatment [2, 2]. Assessments employed diverse neuropsychological batteries evaluating verbal learning and memory, verbal fluency, expressive and receptive language, visuospatial memory, working memory, attention, executive function, processing speed, mental flexibility, verbal and visual reasoning, and motor coordination [1]. Four studies (33.3%) utilized International Cognition and Cancer Task Force (ICCTF) criteria to identify CRCI through neuropsychological testing [1]. The ICCTF criteria define impairment based on standardized deviations from normative means [2, 4].

Assessments were conducted at multiple timepoints relative to treatment, including before, during, and after therapy [4]. Cognitive decline was most pronounced shortly after chemotherapy, with partial recovery observed within one year in some patients [2]. However, substantial heterogeneity in study design and cognitive measures complicated interpretation of temporal patterns [2].

Impact on Quality of Life and Functional Status

CRCI significantly impacted quality of life in lung cancer survivors [1], though specific quality of life measures and scores were not reported in the lung cancer literature [1, 3, 4]. Associated psychological factors included loneliness, fatigue, anxiety, and depression [3], which appeared to potentiate the negative effects of even mild cognitive impairment [4]. Functional impairments extended to activities of daily living and instrumental activities of daily living [3], though specific data on social reintegration outcomes such as return to work or maintenance of social relationships were not available for lung cancer populations [1, 3, 4].

In the broader cancer survivor literature, cognitive difficulties negatively affected autonomy, return to work, social relationships, and self-confidence [2]. The evidence did not include assessments of caregiver burden [1, 3, 4], representing an important gap in understanding the full societal impact of CRCI.

Mechanistic Insights

Neuroinflammation and Immune System Involvement

Chronic neuroinflammation mediated by the active innate immune system emerged as a central mechanism in CRCI pathogenesis [4]. This process involves increased levels of pro-inflammatory cytokines including IL-1, IL-6, and TNF-alpha, along with elevated cytokine receptors [2, 4]. Activation of microglia leads to neuroinflammation, oxidative stress, and neuronal loss [4]. Immune dysregulation was further evidenced by reduced T-lymphocyte counts [3] and the presence of neuronal autoantibodies in some patients [3].

The inflammatory cascade appears to be initiated by disruption of cellular processes, leading to increased inflammatory components in the bloodstream [4]. Intestinal barrier disruption may contribute to this process through translocation of bacteria or bacterial lipopolysaccharide into the circulation [4]. While these inflammatory mechanisms have been well-characterized in chemotherapy-related cognitive impairment, their specific role in targeted therapy-induced neurotoxicity remains unclear, as none of the sources provided targeted therapy-specific mechanistic data [1, 3, 5].

Blood-Brain Barrier and Vascular Changes

Evidence of blood-brain barrier disruption was identified through both direct and indirect mechanisms. Blood-brain barrier leakage was documented in lung cancer patients with cognitive impairment [3], though the sources did

not specify whether this was related to targeted therapy exposure. Endothelial dysfunction was hypothesized as a contributing mechanism to CRCI pathogenesis [4]. DNA damage was implicated as part of the broader pathogenic process [4], potentially leading to vascular injury.

Neurotransmitter and Metabolic Alterations

Dysregulation in glutamate and glutamate metabolism was identified as a factor associated with cognitive impairment in lung cancer patients [3]. Genetic polymorphisms in neurotransmitter-related genes, including catechol-O-methyltransferase (COMT) and brain-derived neurotrophic factor (BDNF), were associated with predisposition to cognitive impairment [2]. These polymorphisms affect neurotransmitter metabolism and synaptic plasticity [2].

Suppression of neurogenesis, particularly in the hippocampus, represents another mechanism by which cancer treatments may impair cognition [2]. This process is mediated through alterations in neurotransmitter metabolism and involves COMT and BDNF polymorphisms [2].

Mitochondrial Dysfunction and Direct Neurotoxicity

Mitochondrial dysfunction contributes to cognitive deficits in cancer patients [2, 3]. Direct neurotoxic injury from systemic treatment and radiation was documented [4], though the specific neurotoxic mechanisms of targeted therapies remain poorly characterized. In animal models, hippocampus-based memory and executive functions mediated by the frontal lobes were particularly susceptible to chemotherapy effects [2].

Structural and Functional Brain Changes

Neuroimaging studies revealed both structural and functional brain alterations in cancer patients with cognitive impairment. Structural changes included global reductions in grey matter volume associated with worse performance on neuropsychological tests [4], reductions in grey matter density in prefrontal areas [4], and reductions in white matter microstructure [4]. These volume reductions in grey matter correlated with cognitive decline [4].

Functional changes encompassed alterations in brain activation and connectivity [4]. Altered Default Mode Network connectivity was specifically identified in lung cancer patients with cognitive impairment [3, 3]. Neurological biomarkers of dysfunction included decreased NAA/Cho and NAA/myo-inositol ratios, along with elevated serum axonal phosphorylated neurofilament subunit H (pNF-H) [2].

Peripheral biomarkers showing promise for early detection included damage-associated molecular patterns, inflammatory components, circulating microRNAs, exosomes, short-chain fatty acids, amyloid beta, and tau in plasma or cerebrospinal fluid [4]. However, none of these biomarkers have been validated specifically in the context of targeted therapy-induced cognitive impairment in lung cancer patients.

Synthesis

The available evidence reveals a paradox: while CRCI is recognized as prevalent in lung cancer survivors (affecting approximately one-quarter of patients) [1], the specific contribution of targeted therapies remains poorly characterized. Several factors explain this knowledge gap.

First, most mechanistic studies have focused on chemotherapy-induced cognitive impairment in breast cancer populations [2, 2, 2], with limited direct investigation of targeted therapy neurotoxicity in lung cancer. The single review specifically addressing targeted therapies in lung cancer focused on brain metastases management rather than CRCI

in the broader population [5, 5]. This creates uncertainty about whether the neuroinflammatory, blood-brain barrier, and metabolic mechanisms described for chemotherapy [2–4] apply equally to EGFR-TKIs and ALK inhibitors.

Second, the substantial regional heterogeneity in CRCI prevalence—ranging from 15% in Asia to 44% in North America [1]—likely reflects methodological differences rather than true biological variation. Studies using comprehensive neuropsychological batteries identified more than twice the prevalence (36%) compared to screening tools alone (17%) [1]. This assessment-dependent variation complicates comparison across studies and suggests that true prevalence depends critically on how CRCI is defined and measured. The consistent finding that subjective cognitive complaints (reported by up to 50% of breast cancer patients) [2] substantially exceed objective deficits (15–25%) [2] further underscores the importance of assessment methodology.

Third, the temporal course of CRCI in targeted therapy recipients remains undefined. While chemotherapy-related cognitive impairment shows peak severity shortly after treatment with partial recovery within one year [2], targeted therapies are often administered continuously for months to years. Whether prolonged exposure to EGFR-TKIs or ALK inhibitors leads to progressive, stable, or reversible cognitive changes cannot be determined from the available literature.

The mechanisms linking targeted therapies to cognitive dysfunction likely involve overlapping pathways with chemotherapy—including neuroinflammation [2, 4], mitochondrial dysfunction [2, 3], and blood-brain barrier disruption [3]—but may also include therapy-specific mechanisms related to on-target effects in the central nervous system. The finding of altered Default Mode Network connectivity specifically in lung cancer patients [3] suggests that CNS-penetrant targeted agents might directly affect neural circuits, though causality cannot be established from observational studies.

Current Management of CRCI

Assessment and Screening Practices

Patient-reported symptoms are commonly assessed using questionnaires such as the Functional Assessment of Cancer Therapy-Cognitive Function (FACT-Cog) [2, 2, 4], while formal neuropsychological tests remain the standard for detecting objective deficits [2, 4]. Screening tools include the Mini-Mental State Examination (MMSE) and Short Portable Mental Status Questionnaire (SPMSQ) [1]. Subjective assessments also utilize the EORTC QLQ-C30 cognitive function subscale and the Cognitive Perceived Cognitive Impairment subscale [1]. However, no absolute assessment tool has been confirmed for evaluating CRCI in lung cancer survivors specifically [1].

Intervention Strategies

Non-pharmacological interventions represent the first-line approach to CRCI management. Cognitive training and rehabilitation are recommended to improve cognitive performance [2, 4]. Physical activity and mind-body interventions have demonstrated benefits for cognitive function [4], with exercise programs showing some improvement in cognitive complaints [2]. Behavioral interventions include education, cognitive behavior therapy, and cognitive training [2]. A combination of cognitive stimulation and physical activity is considered effective supportive care [1].

Pharmacological interventions are reserved as a last resort when non-pharmacological approaches prove insufficient [4]. Agents that have been studied include CNS stimulants, anti-dementia drugs, and selective-serotonin reuptake inhibitors [4]. However, there is no evidence supporting the use of erythropoietin or methylphenidate for CRCI [2], and no pharmacological agents are currently approved for this indication [2].

Management of brain metastases in EGFR-mutated and ALK-rearranged NSCLC involves neurosurgery for limited intracranial disease in accessible regions, stereotactic radiosurgery for patients not amenable to surgery, and whole

brain radiotherapy reserved for patients with multiple lesions due to neurocognitive sequelae [5]. The superior systemic response rates and progression-free survival achieved with EGFR and ALK TKIs compared to chemotherapy [5] may influence cognitive outcomes indirectly through better disease control, though this remains unproven.

Healthcare System Involvement

A multidisciplinary approach is recommended, with cooperation between oncologists, neurologists, imaging researchers, and neuroscientists [2]. Early detection and referral to onco-geriatricians or neurologists are suggested for elderly patients [2]. Referral to neuropsychologists is recommended when specialized cognitive rehabilitation is needed [2]. Clinical recognition and psychoeducation about CRCI are considered important components of care [1].

Management Gaps and Future Directions

Standardization and Guidelines

Relevant standardized assessment tools with high sensitivity and reliability for CRCI remain to be established [1, 4]. Variations in measurement methods, assessors, and limited training for assessors create barriers to consistent evaluation [1]. Recommendations for prevention, diagnostics, and management from leading cancer organizations are needed [4], as the optimal strategy and intervention for CRCI have not been identified [1].

Limited high-quality evidence guides current management [2], and no evidence-based treatment guidelines exist [1]. There is a lack of high-quality evidence investigating CRCI specifically among lung cancer survivors [1], necessitating more robust clinical trials to identify adequate management strategies [2].

Rehabilitation and Implementation Barriers

Limited availability of effective management strategies represents an unmet patient need [2]. Referral to neuropsychologists is recommended due to the lack of readily available rehabilitation programs [2]. The need for evidence-based interventions and policies to prevent and reduce CRCI prevalence remains unfulfilled [1].

Service Delivery Improvements

The increasing number of cancer survivors provides rationale for systematic establishment of specialized survivorship clinics to address late toxicities including CRCI [4]. Awareness of CRCI among clinicians is essential for appropriate prevention, diagnosis, and management [4]. When considering CRCI as a treatment side effect, clinicians should weigh risk-benefit ratios to avoid overtreatment [4]. Fostering international understanding and access to established protocols for preventive strategies would improve care delivery [1].

Research Priorities

Future research should focus on understanding the underlying molecular mechanisms of CRCI and discovering novel biomarkers [4]. Studies are needed to define mechanisms and optimize care [2], particularly addressing limitations in current research such as the need for investigations in specific subgroups and better reporting of measurement methods and assessor training [1]. Early treatment interventions to improve long-term quality of life represent an important research direction [4].

References

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